Fibre Frequency Reference Winter Project

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Aim and Context

- GRACE missions require precision measurements between spacecrafts

- This needs to be:
  - Cheap
  - Light weight
  - Resistant to temperature fluctuations
  - Compact

- FFR could satisfy these criteria

- We used various resin materials to package the FFR from temperature fluctuations

Method

- A ~10m arm is contained within three epoxy resins: UR5048 (soft) and ER2218 (medium) , ER2183 (hard)

- For each resin, we recorded:
  - Amplitude w.r.t a sinusoidal temperature
  - Phase w.r.t a sinusoidal temperature

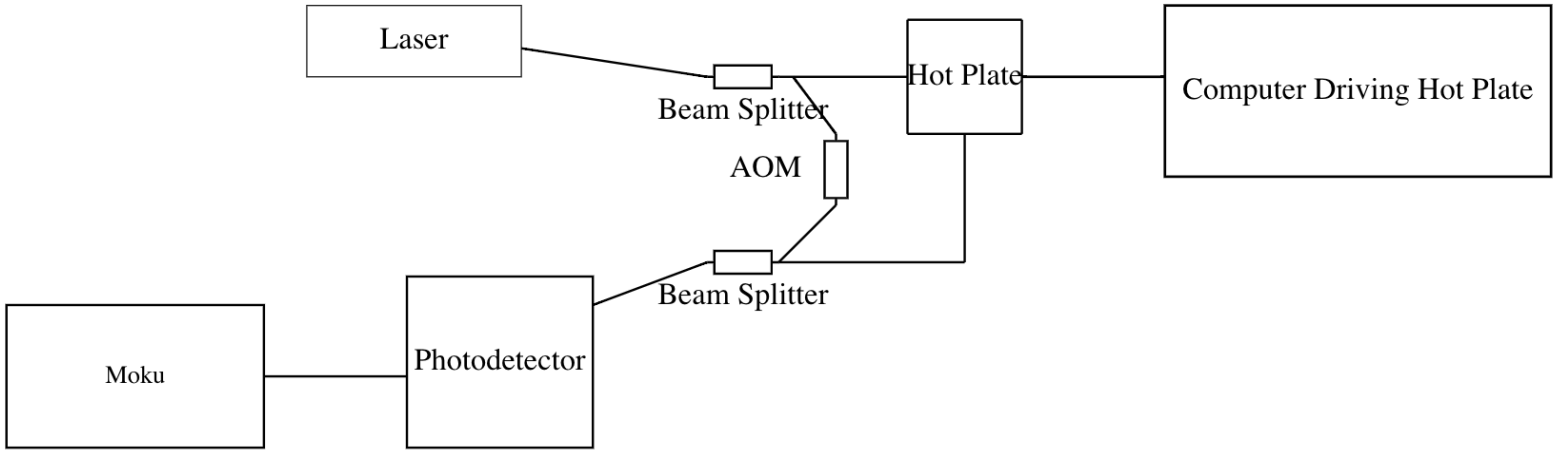


Figure 1: A schematic of the experimental setup that was used for obtaining phase measurement results.

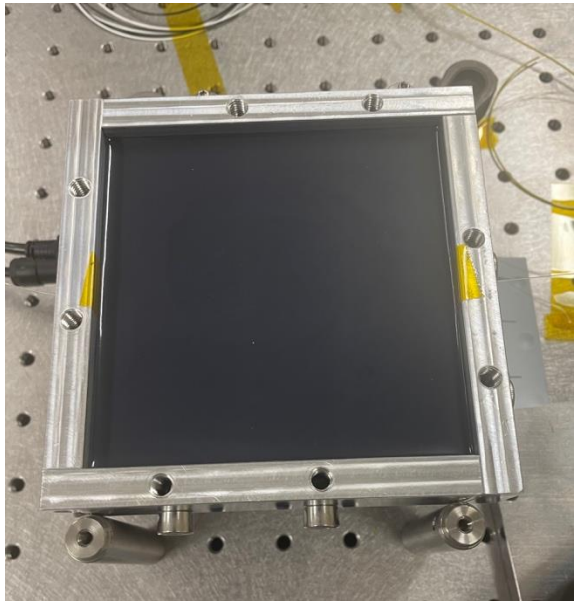


Figure 2: A photo of the FFR arm contained within the hard resin. Note the lid has been removed.

Amplitude Measurements

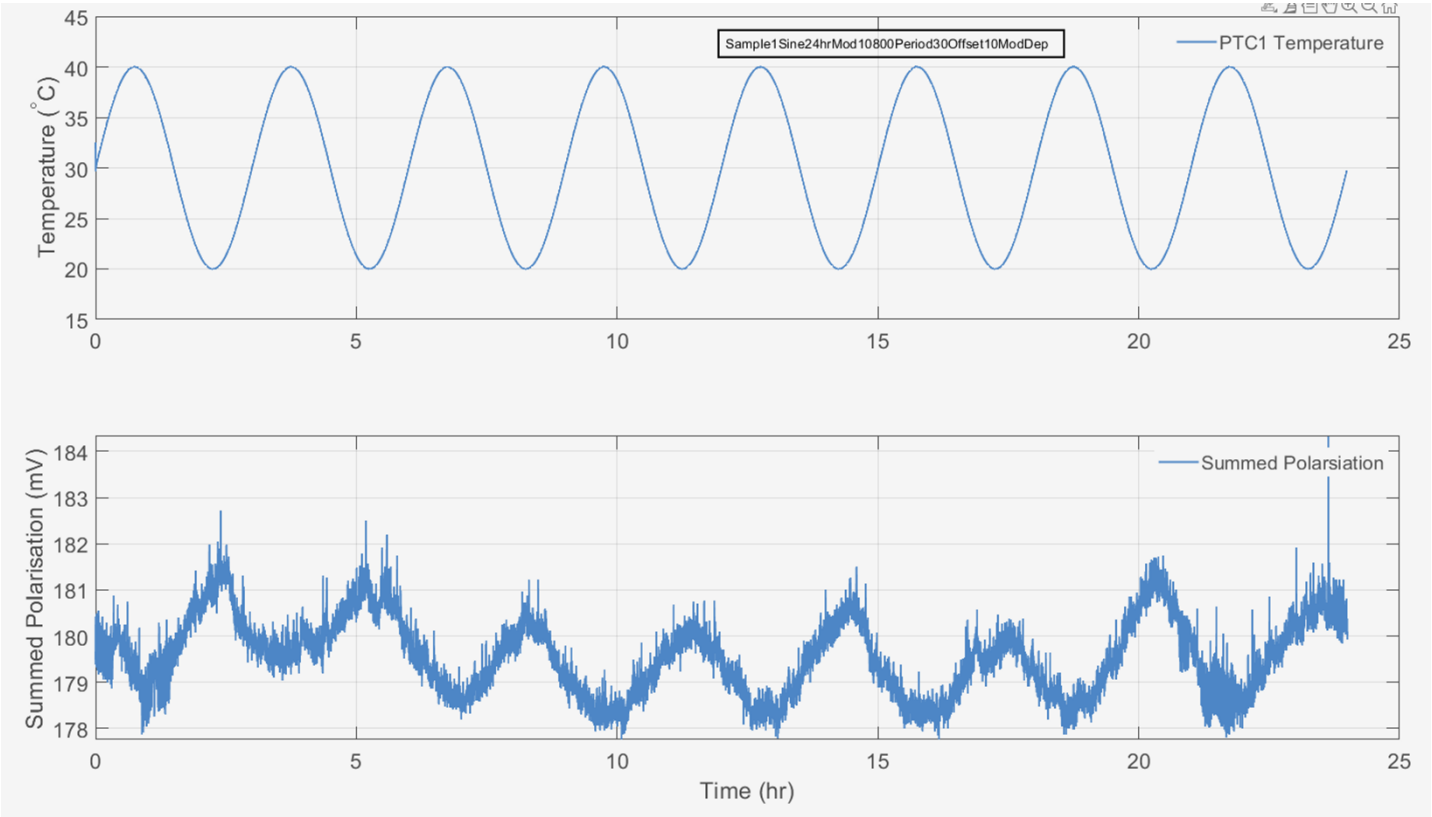


Figure 3: Total Polarization for the soft resin (UR5048) under a 24-hour test with a 180-minute period with 30°C Offset and 10°C Modulation Depth

- Minimal fluctuations in amplitude for UR5048 over the 20°C to 40°C temperature range

Amplitude Measurements

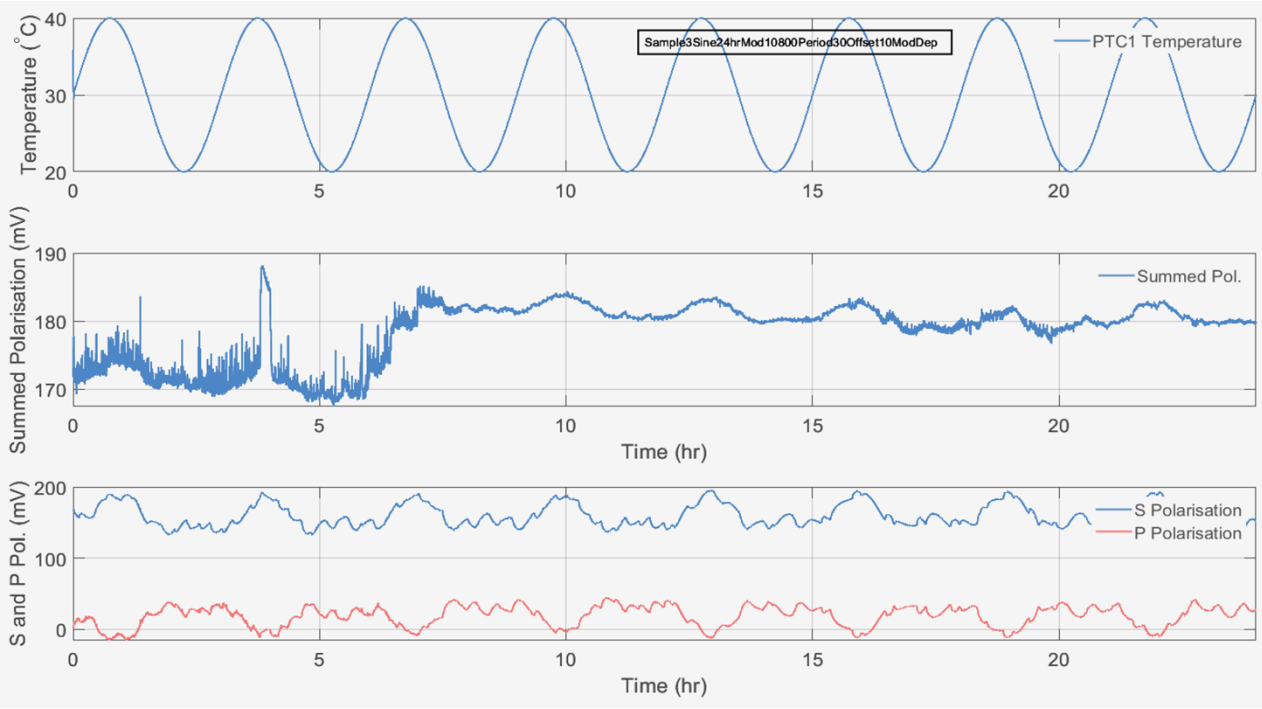


Figure 4 (a): Total polarization along with S and P polarisations for the medium resin (ER2218) under a 24-hour test with a 180-minute period with 30°C Offset and 10°C Modulation Depth

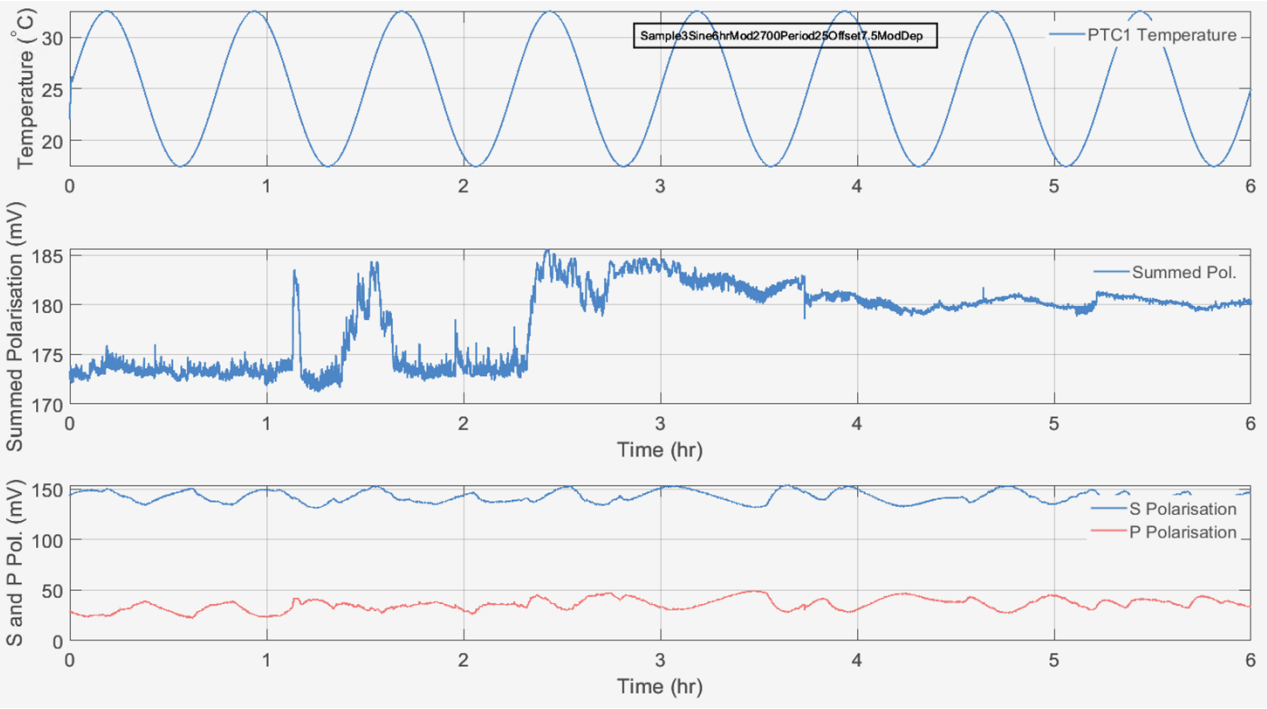


Figure 4 (b): Total polarization along with S and P polarisations for the medium resin (ER2218) under a 6-hour test with a 45-minute period with 25°C Offset and 7.5°C Modulation Depth

- After ~3 periods, there is a noticeable jump, perhaps due to stress on the fibre

Phase Measurements

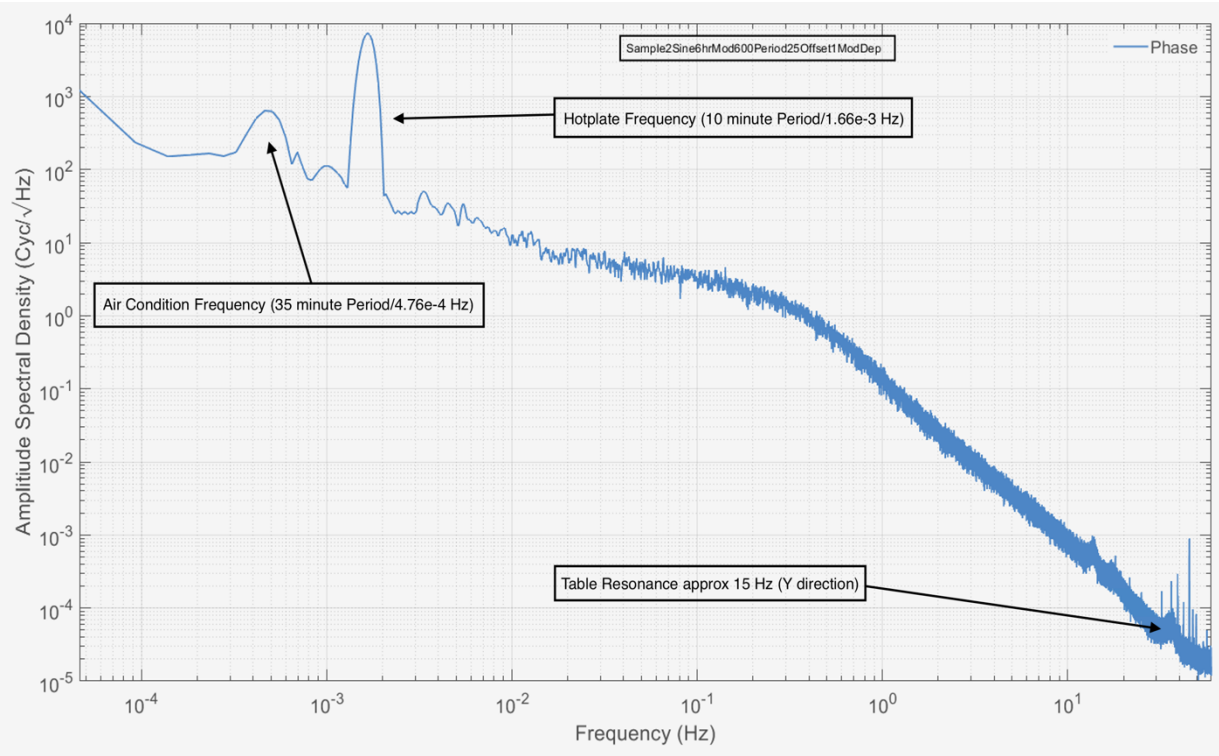


Figure 5 (a): Amplitude Spectral Density for the medium resin (ER2183) under a 6-hour test with a 10-minute period with 25°C Offset and 1°C Modulation Depth

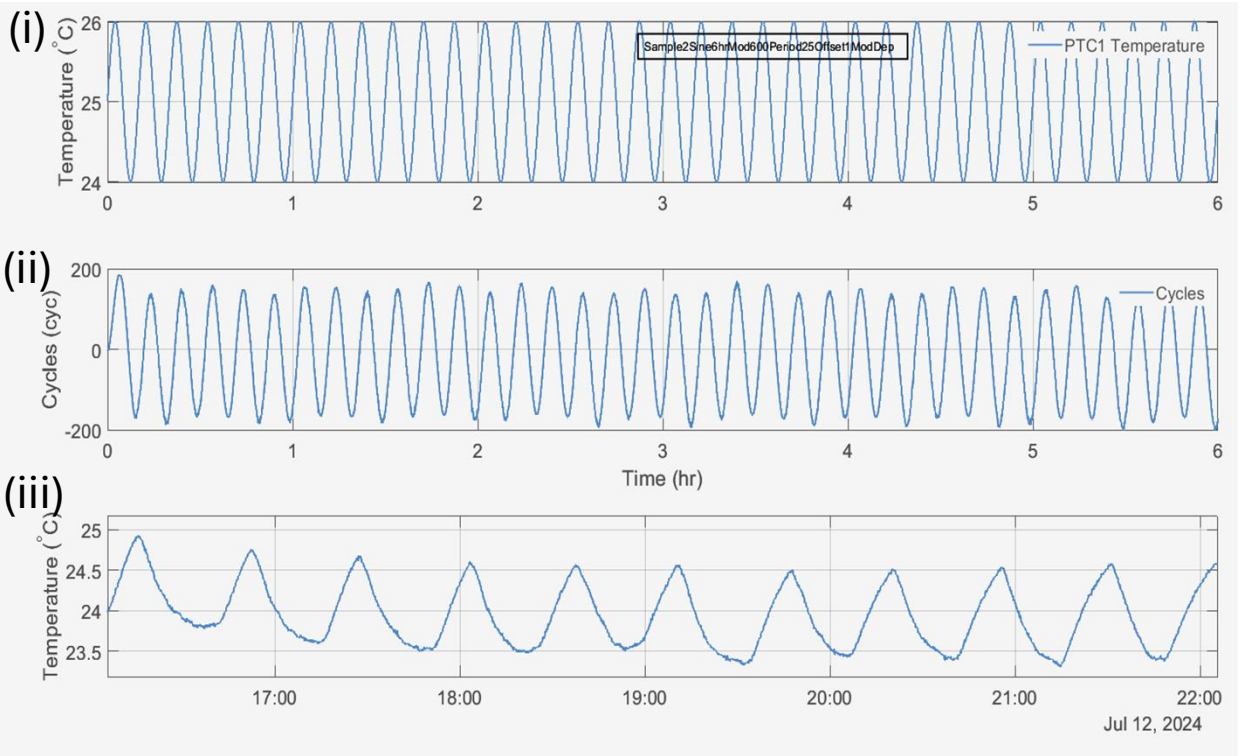


Figure 5 (b): (i) Hot plate temperature, (ii) Cycles, (iii) Lab temperature, variation due to air conditioning

Conclusions

- The soft resin (UR5048) shows great resistance to thermal fluctuations from both amplitude and phase measurements

- The harder resins (ER2218 and ER2183) appears to place stress on fibre after several temperature periods and more prone to temperature fluctuations.